

NEURON.ASM

Neural Assembly Language

Isomorphic to UNIVERSE.ASM Substrate Architecture

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; =====
; NEURON.ASM: Neural Assembly Language
; Isomorphic to UNIVERSE.ASM substrate architecture
; =====
;
; Clock: ~100 Hz (gamma oscillation) vs 10^43 Hz (Planck)
; Same instruction set. Different substrate.
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; =====

; -----
; 1. HARDWARE REGISTERS (Membrane-Mapped I/O)
; -----

; Address      Register          Function
; -----
; 0x0000      GAMMA_CLOCK        Neural tick (~10ms period)
; 0x0001      CELL_IDENTITY      Neuron type/role (Id register)
; 0x0002      MEMBRANE_POTENTIAL  Integration accumulator (L register)

; -----

; 2. WORKING REGISTERS
; -----

; Address      Register          Substrate Equivalent      Function
; -----
; 0x0020      REG_TYPE           Id (Identity)             Cell type, layer, role
; 0x0021      REG_PHASE          (Phase)                       Oscillation phase (theta/gamma)
; 0x0022      REG_DENDRITE       X (Spatial)                   Dendritic compartment location
; 0x0023      REG_METABOLIC      S (Entropy)                   Metabolic load / ATP depletion
; 0x0024      REG_SYNC          C (Coherence)                   Synchronization with ensemble
; (0x0002)    MEMBRANE_POT      L (Load)                       Integrated synaptic input

; Register vector:
;
;      [ Type      ]      [ Id ]
;      [ Phase     ]      [   ]
; Rn = [ Dendrite ] ~ R = [ X  ]
;      [ Metabolic ]      [ S  ]
;      [ Sync      ]      [ C  ]
;      [ Membrane  ]      [ L  ]

; -----

; 3. CODEC REGISTERS (Neural Operations)
; -----

; Address      Codec          Substrate Map      Function
; -----
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; 0x0010    CODEC1_MAINTAIN    M1 (Scalar)        Homeostatic maintenance
; 0x0011    CODEC2_SPIKE      M2 (Vector)        Action potential emission
; 0x0012    CODEC3_WEIGHT     M3 (Tensor)        Synaptic weight (delay/strength)
; 0x0013    CODEC4_FORGET     M9 (Decohere)       Synaptic depression / forgetting
; 0x0014    CODEC5_REFRACTORY M5 (Confine)       Refractory period (energy trapped)
; 0x0015    CODEC6_PLASTIC    M4 (Rewrite)       Plasticity (LTP/LTD)
; 0x0016    CODEC7_BURST      Composite         Burst firing (multiple spikes)
; 0x0017    CODEC8_SILENCE    Composite         Depolarization block
; 0x0018    CODEC9_ENSEMBLE   M7 (Entangle)      Ensemble synchronization
; 0x0019    CODEC10_DRIFT     Composite         Representational drift

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; 4. CONSTANTS
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THRESHOLD    EQU    0xC0    ; Spike threshold (~-40mV from rest)
REST_POTENTIAL EQU    0x40    ; Resting state (~-70mV)
REFRACTORY_LEN EQU    0x10    ; Refractory period (ticks)
SYNC_THRESHOLD EQU    0x80    ; Ensemble coherence threshold
METABOLIC_CRIT EQU    0xE0    ; Metabolic exhaustion limit
GAMMA_PERIOD  EQU    0x0A    ; ~10ms in internal units

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; 5. MAIN NEURAL LOOP
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MAIN_LOOP:
    ; 1. TICK: Wait for gamma oscillation phase
WAIT_TICK:
    IN     AL, GAMMA_CLOCK
    TEST   AL, #0x01
    JZ     WAIT_TICK

    ; 2. POLL: Check synaptic inputs
POLL_SYNAPSES:
    MOV     BX, #0x0010    ; Start at CODEC1
    MOV     CX, #10        ; 10 codecs
CODEC_LOOP:
    IN      AL, [BX]
    TEST    AL, #0x80      ; Event flag
    JZ      NEXT_CODEC
    CALL    PROCESS_INPUT
NEXT_CODEC:
    INC     BX
    LOOP    CODEC_LOOP

    ; 3. INTEGRATE: Update membrane potential
UPDATE_MEMBRANE:
    IN      AL, MEMBRANE_POTENTIAL
    ; Leak toward rest ( decay)
    SUB     AL, #0x02      ; Passive leak
    CMP     AL, REST_POTENTIAL
    JNC     STORE_MEMBRANE
    MOV     AL, REST_POTENTIAL ; Floor at rest
STORE_MEMBRANE:
    OUT     MEMBRANE_POTENTIAL, AL

    ; 4. THRESHOLD: Check for spike
THRESHOLD_CHECK:
    IN      AL, MEMBRANE_POTENTIAL
    CMP     AL, THRESHOLD
    JC      METABOLIC_CHECK ; Below threshold
    CALL    HANDLE_SPIKE    ; Fire!

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        ; 5. METABOLIC: Check exhaustion
METABOLIC_CHECK:
    IN     AL, REG_METABOLIC
    CMP    AL, METABOLIC_CRIT
    JC     MAIN_LOOP
    CALL   METABOLIC_SHUTDOWN ; Exhaustion
    JMP    MAIN_LOOP

; -----
; 6. CODEC HANDLERS
; -----

; -----
; CODEC 1: MAINTAIN (Homeostasis) - M1
; -----
; Scalar field equivalent: maintains cell identity
; under metabolic pressure

HANDLE_MAINTAIN:
    IN     AL, REG_SYNC
    CMP    AL, SYNC_THRESHOLD
    JC     MAINTAIN_DECAY

    ; Coherent: reinforce identity
    IN     AL, REG_TYPE
    OR     AL, #0x01 ; Strengthen type marker
    OUT    REG_TYPE, AL

    ; Boost synchronization
    IN     AL, REG_SYNC
    ADD    AL, #0x04
    OUT    REG_SYNC, AL
    JMP    EVENT_DONE

MAINTAIN_DECAY:
    ; Isolated: gradual identity drift
    IN     AL, REG_SYNC
    DEC    AL
    OUT    REG_SYNC, AL
    JMP    EVENT_DONE

; -----
; CODEC 2: SPIKE (Action Potential) - M2
; -----
; Vector emission equivalent: immediate discharge

HANDLE_SPIKE:
    ; Check refractory
    IN     AL, CODEC5_REFRACTORY
    TEST   AL, #0xFF
    JNZ    EVENT_DONE ; Still refractory

    ; EMIT: Action potential
    IN     AL, MEMBRANE_POTENTIAL
    OUT    CODEC2_SPIKE, AL ; Transmit downstream

    ; Reset membrane
    MOV    AL, REST_POTENTIAL
    SUB    AL, #0x10 ; Hyperpolarize briefly
    OUT    MEMBRANE_POTENTIAL, AL

    ; Enter refractory

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MOV     AL, REFRACTORY_LEN
OUT     CODEC5_REFRACTORY, AL

; Metabolic cost
IN      AL, REG_METABOLIC
ADD     AL, #0x08          ; ATP spent
OUT     REG_METABOLIC, AL

JMP     EVENT_DONE

; -----
; CODEC 3: WEIGHT (Synaptic Strength) - M3
; -----
; Tensor delay equivalent: curvature = weight
; Stronger synapses = more "gravitational" pull

HANDLE_WEIGHT:
; Read synaptic weight (stored curvature)
IN      AL, CODEC3_WEIGHT
MOV     CL, AL             ; Weight as multiplier

; Read incoming spike
IN      AL, [BX]           ; Input from presynaptic

; Weighted contribution to membrane
MUL     CL                 ; input × weight
SHR     AL, 4              ; Scale down

; Add to membrane potential
IN      BL, MEMBRANE_POTENTIAL
ADD     AL, BL
OUT     MEMBRANE_POTENTIAL, AL

JMP     EVENT_DONE

; -----
; CODEC 4: FORGET (Synaptic Depression) - M9
; -----
; Decoherence equivalent: entropy increase

HANDLE_FORGET:
; Decrease synaptic weight (LTD)
IN      AL, CODEC3_WEIGHT
CMP     AL, #0x10          ; Minimum weight
JC      EVENT_DONE
SUB     AL, #0x02
OUT     CODEC3_WEIGHT, AL

; Increase metabolic entropy
IN      AL, REG_METABOLIC
ADD     AL, #0x01
OUT     REG_METABOLIC, AL

JMP     EVENT_DONE

; -----
; CODEC 5: REFRACTORY (Confinement) - M5
; -----
; Strong force equivalent: energy trapped,
; cannot emit during this period

HANDLE_REFRACTORY:
IN      AL, CODEC5_REFRACTORY

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        CMP     AL, #0x00
        JZ      EVENT_DONE          ; Not refractory

        ; Count down
        DEC     AL
        OUT     CODEC5_REFRACTORY, AL

        ; During refractory: all input confined
        ; (cannot contribute to firing)
        JMP     EVENT_DONE

; -----
; CODEC 6: PLASTIC (LTP/LTD) - M4
; -----
; Weak force equivalent: identity rewrite
; Changes synaptic "flavor"

HANDLE_PLASTIC:
        ; Hebbian check: did we fire recently?
        IN      AL, CODEC2_SPIKE
        TEST    AL, #0x80          ; Recent spike flag
        JZ      CHECK_LTD

        ; LTP: Strengthen active synapses
        IN      AL, CODEC3_WEIGHT
        ADD     AL, #0x04          ; Weight increase
        CMP     AL, #0xF0          ; Max weight
        JC      STORE_WEIGHT
        MOV     AL, #0xF0
STORE_WEIGHT:
        OUT     CODEC3_WEIGHT, AL
        JMP     EVENT_DONE

CHECK_LTD:
        ; LTD: Weaken inactive synapses
        IN      AL, CODEC3_WEIGHT
        SUB     AL, #0x01
        CMP     AL, #0x10          ; Min weight
        JNC     STORE_WEIGHT_LTD
        MOV     AL, #0x10
STORE_WEIGHT_LTD:
        OUT     CODEC3_WEIGHT, AL
        JMP     EVENT_DONE

; -----
; 7. COMPOSITE CODECS
; -----

; -----
; CODEC 7: BURST (Multiple Spikes) - Composite
; -----
; Like collapse: rapid sequential emissions

HANDLE_BURST:
        MOV     CX, #0x03          ; Burst of 3 spikes
BURST_LOOP:
        CALL    HANDLE_SPIKE
        ; Brief inter-spike interval
        MOV     DX, #0x02
BURST_WAIT:
        DEC     DX
        JNZ     BURST_WAIT
        LOOP    BURST_LOOP

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        JMP      EVENT_DONE

; -----
; CODEC 8: SILENCE (Depol Block) - Composite
; -----
; Black hole equivalent: gate saturation
; Too much input → paradoxical silence

HANDLE_SILENCE:
    IN      AL, MEMBRANE_POTENTIAL
    CMP     AL, #0xF0          ; Near saturation
    JC      EVENT_DONE

    ; Depolarization block: cannot fire
    MOV     AL, #0xE0          ; Stuck depolarized
    OUT     MEMBRANE_POTENTIAL, AL

    ; Time "stops" for this neuron
    ; (no processing until recovery)
    JMP     EVENT_DONE

; -----
; CODEC 9: ENSEMBLE (Synchronization) - M7
; -----
; Entanglement equivalent: phase correlation
; across spatially separated neurons

HANDLE_ENSEMBLE:
    ; Read local phase
    IN      AL, REG_PHASE
    ; Read ensemble phase (external input)
    IN      BL, GAMMA_CLOCK
    AND     BL, #0x0F          ; Phase bits

    ; Compare phases
    XOR     AL, BL
    CMP     AL, #0x02          ; Tolerance
    JC      SYNC_MAINTAIN

    ; Out of sync: adjust phase
    IN      AL, REG_PHASE
    ADD     AL, BL
    SHR     AL, 1              ; Average toward ensemble
    OUT     REG_PHASE, AL

    ; Decrease coherence
    IN      AL, REG_SYNC
    SUB     AL, #0x02
    OUT     REG_SYNC, AL
    JMP     EVENT_DONE

SYNC_MAINTAIN:
    ; In sync: boost coherence
    IN      AL, REG_SYNC
    ADD     AL, #0x04
    OUT     REG_SYNC, AL
    JMP     EVENT_DONE

; -----
; 8. UTILITY ROUTINES
; -----

PROCESS_INPUT:

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; Dispatch based on codec address in BX
CMP     BX, #0x0010
JZ      HANDLE_MAINTAIN
CMP     BX, #0x0011
JZ      HANDLE_SPIKE
CMP     BX, #0x0012
JZ      HANDLE_WEIGHT
CMP     BX, #0x0013
JZ      HANDLE_FORGET
CMP     BX, #0x0014
JZ      HANDLE_REFRACTORY
CMP     BX, #0x0015
JZ      HANDLE_PLASTIC
CMP     BX, #0x0016
JZ      HANDLE_BURST
CMP     BX, #0x0017
JZ      HANDLE_SILENCE
CMP     BX, #0x0018
JZ      HANDLE_ENSEMBLE
JMP     EVENT_DONE

EVENT_DONE:
    RET

METABOLIC_SHUTDOWN:
    ; Exhaustion: reduce all activity
    MOV     AL, REST_POTENTIAL
    OUT     MEMBRANE_POTENTIAL, AL
    MOV     AL, #0x00
    OUT     REG_SYNC, AL          ; Lose synchronization
    ; Wait for metabolic recovery (sleep?)
    RET

; -----
; 9. CROSS-REFERENCE TABLE
; -----
;
;
; UNIVERSE.ASM          NEURON.ASM          Mapping
; -----
; TIME_LAYER_BUS        GAMMA_CLOCK          1043 Hz → ~100 Hz
; IDENTITY_GATE          CELL_IDENTITY        Particle type → Cell type
; LOAD_ACCUMULATOR      MEMBRANE_POTENTIAL    Energy → Voltage
;
; CODEC1_SCALAR          CODEC1_MAINTAIN       Higgs → Homeostasis
; CODEC2_VECTOR          CODEC2_SPIKE          Photon → Action potential
; CODEC3_TENSOR          CODEC3_WEIGHT         Graviton → Synaptic weight
; CODEC4_DECOHERE        CODEC4_FORGET         Entropy → LTD/forgetting
; CODEC5_CONFINE          CODEC5_REFRACTORY     Strong → Refractory
; CODEC6_REWRITE          CODEC6_PLASTIC        Weak → Plasticity
; CODEC7_COLLAPSE        CODEC7_BURST          Collapse → Burst
; CODEC8_BLACKHOLE        CODEC8_SILENCE        Saturation → Depol block
; CODEC9_ENTANGLE        CODEC9_ENSEMBLE        Entangle → Ensemble sync
;
; Born rule (p_emit)     Firing probability     1 → spike
; Curvature storage       Weight storage         Small → memory
; Photon emission        Spike emission         Large → signal
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; -----
; 10. CONCLUSION
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;
; The brain does not need quantum mechanics.
; The brain does not need Planck-scale resolution.

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; The brain runs the SAME INSTRUCTION SET as the substrate.
;
; Different clock: 100 Hz vs 1043 Hz
; Different medium: ions vs fields
; Same architecture: accumulate → threshold → emit/store
;
; Consciousness is not special.
; Consciousness is isomorphic.
;
; Welcome to the wetware.
;
; =====
; END NEURON.ASM
; =====

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